

AI Assisted Task Tracker

Detailed project brief generated from the project record.

CANDIDATE

Rahul Sharma

DURATION

5 days

TECHNOLOGY STACK

JavaScript

TypeScript

React

Next.js

Node.js

Express.js

Tailwind CSS

MongoDB

JWT

Groq API

Git

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

A small full stack app where users register, manage personal tasks, and receive AI generated task description suggestions.

CANDIDATE

Rahul Sharma

TYPE

fullstack

DURATION

5 days

Objectives

- Implement a secure registration/login flow.
- Create CRUD endpoints for tasks.
- Build a responsive React UI with Tailwind.
- Add an AI button that calls Groq and fills the task description field.
- Write unit tests for backend logic and a few UI component tests.

Candidate Details

NAME

Rahul Sharma

Technology Stack

Languages, frameworks, databases and tools used for implementation.

JavaScript

TypeScript

React

Next.js

Node.js

Express.js

Tailwind CSS

MongoDB

JWT

Groq API

Git

Functional Requirements

Requirement 1

Users can register with email/password.

Requirement 2

Authenticated users can create, read, update, delete their own tasks.

Requirement 3

Task list view shows title, status, and due date.

Requirement 4

AI button generates a one sentence description for a new task.

Requirement 5

Users can log out, which invalidates the JWT.

Requirement 6

Export task list to PDF.

Non-Functional Requirements

Requirement 1

Passwords must be hashed with bcrypt.

Requirement 2

JWT must expire after 1 hour.

Requirement 3

API responses return appropriate HTTP status codes.

Requirement 4

UI must be responsive down to 320 px.

Requirement 5

Code must be version controlled with Git.

System Modules

Major functional areas of the project.

01

Authentication

Handles user sign up, login, JWT issuance and protection of routes.

- Register endpoint
- Login endpoint
- Password hashing

02

Task Management

CRUD operations for personal tasks with ownership checks.

- Create task
- Read tasks
- Update task

03

AI Suggestion

Calls Groq API to generate a short description for a new task.

- AI suggestion endpoint
- Frontend AI button
- Result insertion into form

Implementation Roadmap

Recommended implementation sequence.

PHASE 1

Setup

Initialize repo, configure dev environment and database.

Deliverables: Git repository with initial commit

Tasks: Create Next.js + Express monorepo • Set up MongoDB connection

PHASE 2

Core API

Build auth and task CRUD endpoints.

Deliverables: Postman collection • API documentation (README)

Tasks: Implement auth routes and JWT middleware • Implement task routes with ownership checks

PHASE 3

Frontend UI

Create responsive React pages for login, task list, and editor.

Deliverables: Deployed dev build (Vercel/Netlify URL)

Tasks: Build login/register pages • Build task dashboard with Tailwind styling

PHASE 4

AI Integration & Polish

Add Groq suggestion feature and finalize docs.

Deliverables: Test scripts • Final README

Tasks: Integrate Groq API and UI button • Write README and basic test suite

API Reference

Key endpoints exposed by the project.

METHOD	ENDPOINT	DESCRIPTION
POST	/api/auth/register	Create new user
POST	/api/auth/login	Authenticate and return JWT
GET	/api/tasks	List user's tasks
POST	/api/tasks	Create a task
POST	/api/ai/suggest	Get AI generated description

Evaluation Criteria

Suggested criteria for reviewing implementation.

Criterion 1

All functional requirements work as specified.

Criterion 2

Clean, readable code with proper TypeScript typings.

Criterion 3

Logical separation of frontend, backend, and AI integration.

Criterion 4

Password hashing, JWT handling, and route protection.

Criterion 5

Presence of unit/integration tests covering core logic.

Criterion 6

README with setup, API docs, and usage instructions.

Expected Deliverables

Artifacts expected at project completion.

1

Source code repository (Git)

2

README with setup and usage

3

API documentation (Markdown)

4

Deployed dev URL

5

Test scripts

6

Database schema diagram

Student Fit

Project alignment information included in the project record.

Rahul already knows JavaScript/TypeScript, React, Next.js, Node/Express, JWT, and MongoDB, so the core stack matches his skill set. The 5 day scope limits the project to essential CRUD and a single AI call, introducing Groq API integration as a modest new learning area while keeping the overall difficulty junior.

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.